Syllabus for AI/ML:

Data Analytics:

- CENTRAL TENDENCY:
  - Mean
  - Median
  - Mode
  - Variance
  - Standard Deviation (SD)
  - Coefficient of Variation (CV)
  - Covariance
  - Correlation Coefficient
  - Probability
  - Probability Distributions
    - Discrete
    - Continuous
    - Binomial

- MISSING PARTS:
  - Linear Algebra
    - Vectors & Matrices
    - Matrix multiplication
    - Determinant
    - Eigenvalues & Eigenvectors
    - Dot product
    - Norms

Machine Learning:

- FOR CODING:
  - COMPULSORY:
    - Numpy – For Data Cleaning
    - Pandas – For Dataset Handling
    - Matplotlib – For Visualization

  - DATA CLEANING CONCEPTS:
    - Handling missing values
    - Handling duplicate values

  - DATA PREPROCESSING:
    - Scaling
    - Encoding

- REGRESSION **(Given in pdf)**
  - Simple Linear Regression
  - Multiple Linear Regression
  - Mse In Regression(Mean Square Error)
  - Ridge Regression
  - Lasso Regression

- CLASSIFICATION **(Given in pdf)**
  - Random Forest Classification
  - Random Forest Classifier Using Scikit-Learn
  - Knn Classification
  - Decision Tree And Post Pruning Practical Implementation
  - Decision Tree And Pre Pruning Practical Implementation
  - Naive Bayes Classifier
  - Interpret Confusion Matrix Precision, Recall, Accuracy, F1 Beta

- MISSING PARTS:
  - Basic concepts of ML
    - Supervised, Unsupervised, Reinforcement Learning and its respective algorithms
    - When to use which Algorithm for which Dataset
    - Which Metric for which algorithm
    - Logistic regression (When to prefer Linear / Logistic regression over each other)
    - Overfitting vs Underfitting

- Train / Validation / Test Split
- Regularization
    - L1 (Lasso) → removes features
    - L2 (Ridge) → reduces weights
- Confusion Matrix

Deep Learning:

- **FROM PDF**
  - Perceptron Trick
  - Perceptron Trick
  - Handwritten Digit Classification Using Ann
  - Backpropagation
  - Image Classification Using Cnn
  - Rnn Model
  - Epoch
  - Optimizer
  - Adam
  - Dropout Regression
  - Dropout Classification Problem
  - Regularization L2

- Missing Parts:
  - Tensorflow/pytorch for programming
  - Most things from ML
  - Batch, Layers
  - FNN
  - CNN (Computer Vision + DL)
  - RNN

Gen AI

- Natural Language Processing (NLP):
  - Tokenization
  - TF-IDF
  - More concepts exists other than these important ones, research it yourself
- LLM Basics
  - What is an LLM
  - Pretraining vs Fine-tuning
  - Tokens & Context Window
- Transformers
- RAG pipeline (Research in detail from chatgpt)
  - Notes 1
  - Notes 2